Ramadhan Adam Zome

M.Sc. candidate in Artificial Intelligence and Machine Learning · Research exchange, Hiroshima University

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RESEARCH INTERESTS

Open-world and out-of-distribution detection; self-supervised representation learning; few-shot adaptation; machine learning for security, in particular malware analysis at the level of the Windows executable format; evaluation that shows where models fail.

EDUCATION

Pan African University Institute for Basic Sciences, Technology and Innovation (PAUSTI) 2025 – 2027 (expected)
M.Sc. Artificial Intelligence and Machine Learning Nairobi, Kenya

- Thesis: *Open-World Windows Malware Detection with PE-Aware Hierarchical Embeddings and Few-Shot Adaptation*. Supervisors: Dr. Albert Njoroge Kahira and Dr. Mulang' Onando (PAUSTI); Assoc. Prof. Junjun Zheng (Hiroshima University).

Hiroshima University, Dependable Systems Laboratory Aug 2026 – Jan 2027
Special Research Student (research exchange), host: Assoc. Prof. Junjun Zheng Higashi-Hiroshima, Japan

- Pretraining the PE-aware byte encoder on SOREL-20M with the laboratory's GPUs, evaluating open-set rejection on BODMAS, and presenting the work at the Chugoku-section joint conference (October 2026).

Kenyatta University 2019 – 2023
B.Sc. Statistics and Programming, First Class Honours Nairobi, Kenya

PRESENTATIONS

- **R. A. Zome**, J. Zheng, H. Okamura, T. Dohi. *Open-World Windows Malware Detection with PE-Aware Hierarchical Embeddings and Few-Shot Adaptation*. 2026 (77th) Joint Conference of Electrical and Information Engineers, Chugoku Section, Session 25-1, Hiroshima University, 17 October 2026. Oral presentation.

RESEARCH EXPERIENCE

M.Sc. thesis research: open-world Windows malware detection 2025 – present
PAUSTI and Hiroshima University

- **Question.** Can an encoder trained without labels recognise known malware families, reject families it has never seen, and later learn a new family from a few examples?
- **PE-aware byte encoder.** Designed a hierarchical transformer that reads raw bytes region by region (header, sections, overlay), with a learned structural attention bias over section type, byte entropy and relative size. Pretrained it contrastively on **9.9M SOREL-20M malware binaries** (208K steps, 140 A100-hours).
- **Open-set layer.** Families are described by several HDBSCAN prototypes, distances are Mahalanobis with Ledoit–Wolf shrinkage, and each family has its own threshold. Zero-day AUROC on BODMAS: **0.81** (0.83 on 121 families first seen after the training period), against a target of 0.70.
- **Evidence that pretraining helps.** 5-NN family recall of 0.80 on 102 families, ahead of byte-statistic references on 10 of 10 splits (+0.043); none of the test files appear in the pretraining corpus.
- **Negative result, reported.** A second encoder on 1,862 EMBER2024 features (20M parameters, 3.7M training rows) did not beat its own raw features; ablations showed that rewriting bytes (XOR, instruction substitution) did not improve on removal-only views.
- **Infrastructure.** Fault-tolerant ingestion of 9.9M binaries from S3 into 1,986 memory-mapped shards (process isolation, memory ceilings, atomic writes; 115K files per hour at peak, under 0.7% retried or skipped).

SELECTED PROJECTS

Windows PE internals: raw-pe and sco-pe | NASM x86-64, C github

- Hand-assembled a working 2,560-byte Win64 executable in NASM, every header and import entry written by hand with no linker; wrote a bounds-checked PE parser in C (headers, sections, imports, exports, overlays).

Explainable deepfake detection with retrieval | PyTorch, FAISS, FastAPI live demo

- Xception classifier on 140K real and GAN faces (about 97% held-out accuracy on 20K images) with retrieval-augmented explanations grounded in the deepfake-forensics literature.

Swahili GPT | PyTorch model

- Character-level GPT trained from scratch on 65M characters of Swahili news (29,544 articles), a study of language modelling for a low-resource African language.

Phish-Transformer | PyTorch, Flask github

- 45K-parameter character-level transformer for phishing URLs (test AUC about 0.98), served to a Chrome extension.

PROFESSIONAL EXPERIENCE

Innovative Toll Solutions Dec 2023 – Aug 2025
Software Engineer Nairobi, Kenya

- Modelling and reporting on operational logs and transaction data; log-level anomaly detection, alert triage and traffic-pattern analysis; resilience work after DDoS incidents.

GPU platform pilot evaluation for PAUSTI Aug 2026
Student evaluator, Kinesis Network A100 pilot

- Benchmarked training and inference with a custom language-model workload and wrote the decision report on performance, reliability and data persistence.

WRITING

- Technical articles at medium.com/@ramadhanzome4, including *I Wrote a Windows PE File From Scratch, in Assembly* (2026), *PE Overlays: A Small Thing Worth Writing About* (2026), *Beyond the Chatbot: How Self-Attention Learns Structure by Accident* (2026) and *Row Hammer: How Reading Memory Fast Enough Breaks Memory That Was Never Touched* (2026).

SKILLS

Research	Self-supervised and contrastive learning, open-set recognition, clustering, ablation design, evaluation
Languages	Python, C, SQL, Bash, x86-64 assembly (NASM)
Tools	PyTorch, TensorFlow, scikit-learn, Hugging Face, FAISS, LIEF, LaTeX, Docker, Kubernetes, Git
Security	PE format, static and dynamic malware analysis, reverse engineering (REMnux, FLARE-VM)

SCHOLARSHIPS AND HONOURS

- Pan African University scholarship (African Union), M.Sc. at PAUSTI, 2025 – 2027.
- County government scholarship, Kenya, B.Sc. at Kenyatta University, 2019 – 2023.
- County government scholarship, Kenya, secondary school.
- First Class Honours, B.Sc. Statistics and Programming, Kenyatta University, 2023.

REFERENCES

Dr. Albert Njoroge Kahira (PAUSTI), Dr. Mulang' Onando (PAUSTI), Assoc. Prof. Junjun Zheng (Hiroshima University). Contact details on request.